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therefore, it is important to distinguish between the toxicity of dieldrin and its metabolites. However, limited information is available regarding metabolites formed by the aerobic biodegradation of dieldrin.^{13,16} Therefore, there is still an urgent need to develop an effective method for bioremediation of aerobic zones polluted with organochlorine insecticides. *Phanerochaete chrysosporium* is a commonly studied white rot fungus that showed to degrade and mineralize a wide range of industrial and agricultural pollutants, including DDT.¹⁷ *Phanerochaete chrysosporium* degraded approximately 90% of endosulfan and generated endosulfan sulfate and endosulfan diol as the principal metabolites.¹² Kullman and Matsumura¹² proposed pathways through which the oxidation of diol to hydroxyether was formed and showed that dialdehyde and lactone were end products of the metabolism. Kamei *et al.*¹⁸ also reported that the white rot fungus *Trametes hirsuta* degraded more than 90% of endosulfan and approximately 70% of endosulfan sulfate, producing endosulfan diol and endosulfan dimethylene as metabolites. In other studies, *Trichoderma harzianum*¹⁹ and *Mucor thermohyalospora*²⁰ degraded endosulfan, apparently *via* both the oxidation of endosulfan to endosulfan sulfate and its hydrolysis to endosulfan diol.

The present study reveals soil fungi that can degrade dieldrin and endosulfan and their degradation pathways.

1. Biodegradation of dieldrin

1.1. Degradation experiments

All fungal strains except *Trichoderma* strains, which were used in this study, were grown on PDA agar medium in Petri dishes at 25°C. Fungal disks (6 mm diameter) were taken from the margins of fungal colonies grown for 14 days. Each fungal disk was transferred into 10 mL of modified Czapek-Dox (MCD) liquid medium (Glucose; 10 g, $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$; 0.5 g, NaNO_3 ; 2.0 g, $\text{FeSO}_4 \cdot 4\text{H}_2\text{O}$; 0.01 g, K_2HPO_4 ; 1.0 g, Bacto™ yeast extract (Difco); 0.5 g per liter) in a sterilized 100 mL Erlenmeyer flask. After pre-incubation for 7 days, 50 μL of dieldrin in acetone was added to each inoculated flask (final concentration: 13.2 μM). To prevent volatilization of dieldrin, the flask was sealed with a glass stopper.²¹ The cultures were incubated statically for 14 days at 25°C under dark conditions. As a control, cultures were killed by autoclaving (121°C, 15 min) after 7 days of pre-incubation. Following the incubation period, cultures were homogenized with 15 mL of acetone, after which the residual biomass was removed by centrifugation at 3000 $\times g$ for 10 min. An aliquot of 1 mL of supernatant was then transferred into a test tube and extracted with 5 mL of hexane. Next, dieldrin was analyzed by gas chromatography with a ⁶³Ni electron capture detector (GC/ECD), which was performed on an HP 6890 GC system linked to an HP 5890 detector and a 15 m column (HP50+, J&W Scientific, Folsom, CA, USA). The oven temperature was programmed to increase from 150 to 200°C at 20°C/min and 200 to 280°C at 20°C/min. The recovery rate of dieldrin in the control cultures was found to be 90–100%, which indicated that the extraction was efficient and the analysis deviation was small. In the screening assay there were one and three replicates of samples

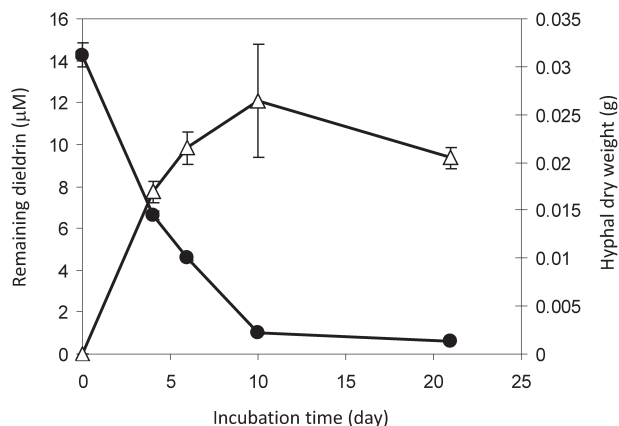


Fig. 1. Time course of dieldrin degradation by strain DDF. Solid circles indicate the concentration of remaining dieldrin and open triangles indicate the hyphal growth as dry weight. Error bars indicate $\pm$ S.E. of triplicate samples.

and controls, respectively. Following screening, the time course degradation experiment for dieldrin-degrading fungus was performed in triplicate.

The degradation experiment was conducted using 35 strains of *Trichoderma* spp. and 36 strains that were isolated from a soil with annual endosulfan applications for more than 10 years. After pre-incubation for 7 days, 50 μL of dieldrin in acetone was added to each inoculated flask (final concentration: 13.2 μM). Among *Trichoderma* spp., strain 93155 was capable of degrading dieldrin with 19.7% (2.6 μM) degradation after 14 days of incubation. However, a fungus isolated from soil contaminated with endosulfan (strain DDF) degraded 95.8% of dieldrin from an initial concentration of 13.2 μM . A subsequent time course degradation experiment revealed strain DDF degraded dieldrin from 14.3 to 1.01 μM during 10-days of incubation at 25°C (Fig. 1).

The ITS rRNA sequence of strain DDF (471 nucleotides, GenBank accession no. AB536702) was compared with those of the fungal sequences in the GenBank database. Strain DDF exhibited a high sequence similarity to those of *Mucoraceae* fungi as indicated by the constructed phylogenetic dendrogram. The highest sequence identity (100%) was found with *Mucor racemosus* f. *racemosus* (GenBank accession no. AY213659); therefore, strain DDF was designated as *Mucor racemosus* strain DDF. These findings are in accordance with those of Anderson *et al.*,⁹ who reported that *Mucor alternans* degraded dieldrin and DDT.

1.2. Detection of the metabolites produced during dieldrin degradation

The dieldrin-degrading fungus was grown in 500 mL of MCD medium containing 13.2 μM dieldrin for 5 days, after which the culture was homogenized and separated into a mycelial fraction and a fluid fraction by centrifugation. The fluid fraction was subsequently acidified to pH 2.0 with 0.1 N HCl and extracted three times with ethyl acetate. The mycelial fraction was also extracted three times with ethyl acetate. The ethyl acetate ex-

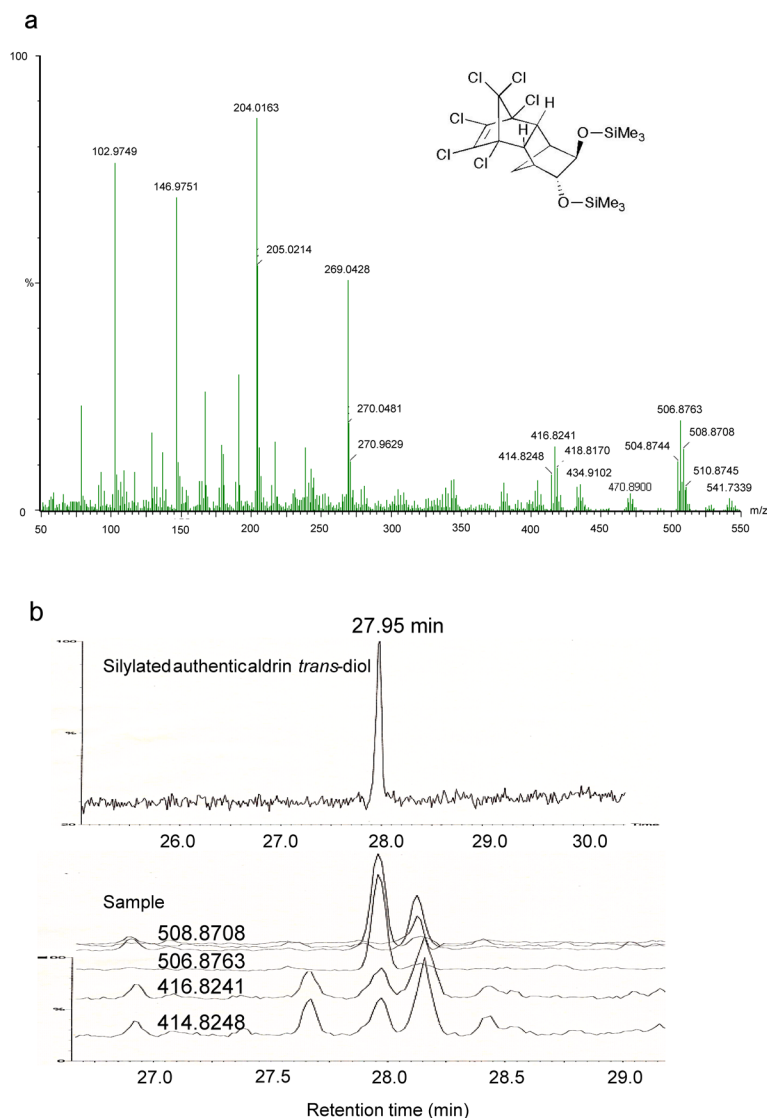


Fig. 2. (a) A mass spectrum of aldrin *trans*-diol derivatized with BSTFA. The base ion peak m/z 73 was omitted because its abundance was too high. Monitoring ions were selected for the comparatively high intensity chlorination pattern. The chemical structure of aldrin *trans*-diol derivatized with BSTFA is also indicated. (b) The retention time of aldrin *trans*-diol derivatized with BSTFA detected by GC/MS analysis with a HRMS-HRGC, and the set of individual intensities of the target ion based on the chlorination pattern.

tracts were then mixed, dried with anhydrous sodium sulfate, and evaporated to dryness. The concentrate was subsequently re-suspended into acetone/hexane (1:1) and derivatized with bis(trimethylsilyl) trifluoroacetamide (BSTFA). Aldrin *trans*-diol in the BSTFA-derivatized concentrate was detected by GC/ECD. The oven temperature was programmed to increase from 150 to 200°C at 20°C/min and then 200 to 280°C at 20°C/min, while the inlet temperature was set at 180°C. Gas chromatography-mass spectrometry (GC/MS) analysis for confirmation of aldrin *trans*-diol was conducted using a HP6890 series high resolution gas chromatograph (HRGC) with a 60 m column (DB5-MS, J&W Scientific, Folsom, CA, USA) interfaced to a high-resolution mass selective detector (HRMS) (Micromass Autospec-Ultima) to detect low concentrations of the material. The oven temperature was programmed to increase from 140 to 200°C at

10°C/min, followed by 200 to 270°C at 2°C/min and then 270 to 300°C at 30°C/min. The inlet temperature was set at 180°C. The full scan mass spectrum of aldrin *trans*-diol derivatized with BSTFA is shown in Fig. 2a. The product generated a molecular ion peak at m/z 542, as well as remarkable fragment ions at m/z 506.8763 and m/z 414.8248. Aldrin *trans*-diol as a metabolite was confirmed by monitoring of the chlorination pattern using the selective ion monitoring (SIM) mode of HRGC-HRMS. The mass spectra m/z 414.8248, 416.8241, 506.8763 and 508.8708 of aldrin *trans*-diol were selected. In addition, a lock-mass m/z (430.9728) from perfluorokerosene (PFK) was used for drift correction. The relative intensities of the target fragment ions of the metabolite were 39.3 (414.8248), 70.2 (416.8241), 100 (506.8763) and 54.0 (508.8708), the metabolite was detected at a retention time of 27.95 min, and its retention time and the relative inten-

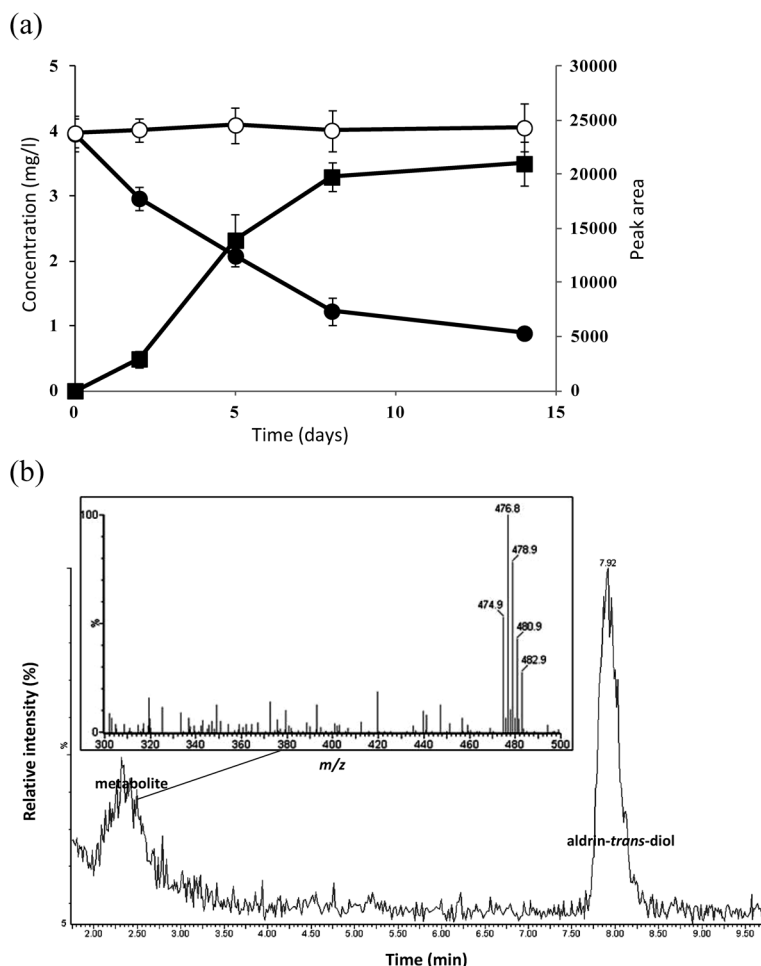


Fig. 3. (a) Time courses for the degradation of aldrin *trans*-diol (5 mg L^{-1}) and peak area responses for the resulting metabolite and (b) mass chromatogram (UPLC-ESI (-)-mode) of *M. racemosus* strain DDF after incubation in liquid medium containing aldrin *trans*-diol.²⁵⁾ Each symbol represents the mean value for triplicate incubations. Solid circles and open circles indicate concentrations of aldrin *trans*-diol in culture with strain DDF and heat-killed control culture, respectively. Solid squares indicate peak area responses for the metabolite ($m/z=476.8$; UPLC-SIM-ESI (-)-mode).

sity of its chlorination pattern matched the standard (Fig. 2b).

A degradation test was performed by adding $50 \mu\text{L}$ of an aldrin *trans*-diol stock solution (1000 mg L^{-1} , dissolved in acetone) to each culture (final aldrin *trans*-diol concentration: 5 mg L^{-1}) following a pre-incubation period. The cultures were acidified to pH 2.0 with 0.1 M aqueous HCl following the degradation test, then extracted with ethyl acetate. The ethyl acetate extracts were evaporated, after which the residues were dissolved in acetonitrile: H_2O (1:1). The presence of any metabolites in the supernatants was determined periodically by UPLC-ESI (-)-MS and HR-FAB (-)-MS.

Inspection of the UPLC-ESI (-)-MS spectra for degradation of aldrin *trans*-diol by strain DDF indicated that the concentration of aldrin *trans*-diol had been reduced and a new peak with an earlier retention time than that of aldrin *trans*-diol had been detected. This new peak had a m/z value of 476.8 in UPLC-ESI (-)-mode (Fig. 3b). Moreover, on the basis of the isotopic fragmentation pattern, the compound possessed six chlorine atoms (Fig. 3b). These findings suggested that the new product

was not formed as a result of an elimination reaction involving the chlorine atoms of the host compound. Additionally, the levels of aldrin *trans*-diol in the liquid culture containing strain DDF were decreased by 78% after 14 days, and the metabolite signal at m/z 476.8 gradually increased with time (Fig. 3a). In an attempt to identify the metabolites, chemical derivatizations of aldrin *trans*-diol were performed. Trial sulfations of the aldrin *trans*-diol were conducted according to standard conditions (SO_3 -pyridine in DMF and/or pyridine at 20 – 60°C); however, the reactions were unsuccessful, possibly because of hydrogen bonding between the two hydroxy groups. Eventually, the reaction was achieved by microwave treatment using the method described by Kiyota *et al.*²²⁾ Briefly, the crude product was purified by reversed-phase TLC to give a 1:1 mixture of the sodium salts of the 6-*exo*- and 7-*endo*-sulfates. This mixture was further purified to give an 83:17 mixture of the 6-*exo*- and 7-*endo* isomers. The stereochemistry of the product was determined by a nuclear Overhauser effect spectroscopy (NOESY) experiment. UPLC-ESI (-)-MS analysis indicated that the retention times

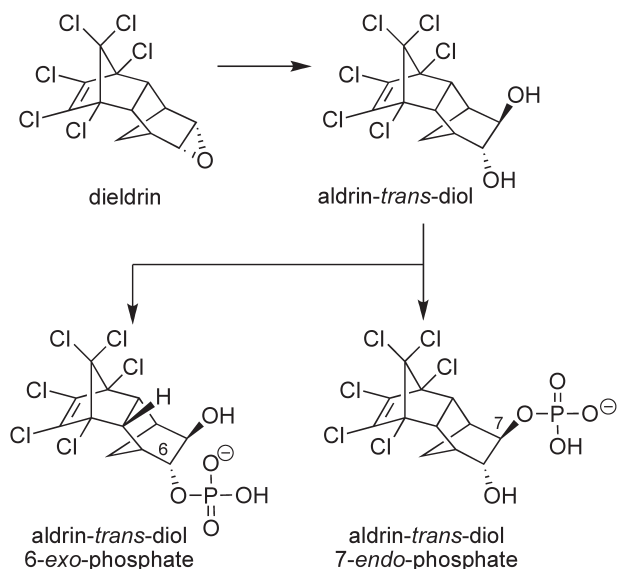


Fig. 4. Proposed metabolic pathway of degradation by *M. racemosus* strain DDF.²⁵⁾

of the synthetic sulfates were 4.32 and 4.71 min in the 6-*exo*- and 7-*endo*-sulfates, respectively. However, the retention time (2.38 min) of the natural metabolite was earlier than that of the synthetic sulfates (4.32 and 4.71 min) (data not shown). Based on these results, the possibility of the metabolites being the corresponding phosphate moiety with the same mass number ($m/z=476.8$ $[M-H]^-$) was considered. Therefore, 6-*exo*-phosphate was prepared by condensation of aldrin *trans*-diol with phosphoric acid using 1-Ethyl-3-(3-dimethylaminopropyl)-carbodiimide (EDCI), after which the stereochemistry was determined by a NOESY experiment. Analysis by HPLC revealed that the retention time of the synthetic phosphorylated aldrin *trans*-diol was in agreement with that of the natural metabolite (data not shown). Furthermore, the natural extract was subjected to HR-FAB (–)-MS analysis, which revealed that the $[M-H]^-$ value (m/z : 474.8400) was very similar to the value calculated for phosphate (m/z : 474.8397 for $C_{12}H_{10}^{35}Cl_6O_5P$), but not that

of sulfate (m/z : 474.8302 for $C_{12}H_9^{35}Cl_6O_5S$). The mass accuracy for the natural metabolite [(measured mass – theoretical exact mass)/theoretical exact mass] was 0.63 ppm when compared with aldrin *trans*-diol phosphate. The identification criterion was based on that of Pihlström,²³⁾ who stipulated a mass accuracy value of less than 5 ppm. Analysis by HR-MS was conducted to evaluate identification of metabolites in the environment²⁴⁾ because the target metabolites were in trace amounts and unstable. When the metabolite was analyzed in UPLC-SIM-ESI (–)-mode using a mobile phase composed of acetonitrile and a 0.2% acetic acid solution (30:70 v/v), the peak signal for the metabolite consisted of two resolved peaks (data not shown). These results indicated that the identities of the natural metabolites were aldrin *trans*-diol 6-*exo*- and 7-*endo*-phosphates. The overall proposed pathway of degradation by strain DDF is shown in Fig. 4.

2. Biodegradation of endosulfan

2.1. Microorganisms and isolation of fungi from soil

Soil dilution (soils A, B and C) and soil washing (soil D) were conducted to isolate the soil fungi from the soil as follows: 1 g of each soil sample was mixed with 20 mL of sterile distilled water using a Universal Homogenizer (Nihon Seiki Seisakusho, Tokyo, Japan), then subjected to 10^{-2} , 10^{-3} and 10^{-4} dilutions. Aliquots of 100 μ L of the resulting soil solutions were subsequently placed onto a plate of Martin agar medium containing chloramphenicol (0.25 g L⁻¹), after which the plates were incubated at 25°C for 5 days. The fungal colonies were subsequently classified into several groups according to their morphology, growth rate and color, after which representatives of each group were picked up with a sterile needle and sub-cultured on potato dextrose agar (PDA; Difco Laboratories, Detroit, MI, USA). Soil washing was then conducted according to Gams and Domsch.¹⁹⁾ Briefly, a 10 g aliquot of soil sample was placed on the rough sieve of the sterilized soil washing apparatus, 90 mL of sterilized water were added and the apparatus was sealed. The apparatus was subsequently shaken vertically for 1 min, after which the water suspension was discarded through the drain tube (washing procedure). The washing procedure was repeated 20 times. Soil ag-

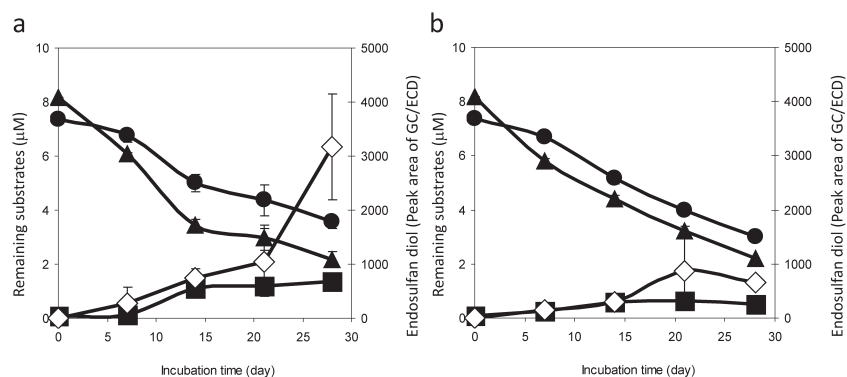


Fig. 5. Time courses of degradation of α - and β -endosulfan by strains W8 (a) and Cm1-45 (b). Solid triangles, circles and squares indicate the concentrations of remaining α -endosulfan, β -endosulfan and endosulfan sulfate, respectively. Open diamonds indicate endosulfan diol. Error bars indicate $\pm$ S.E. for triplicate samples.

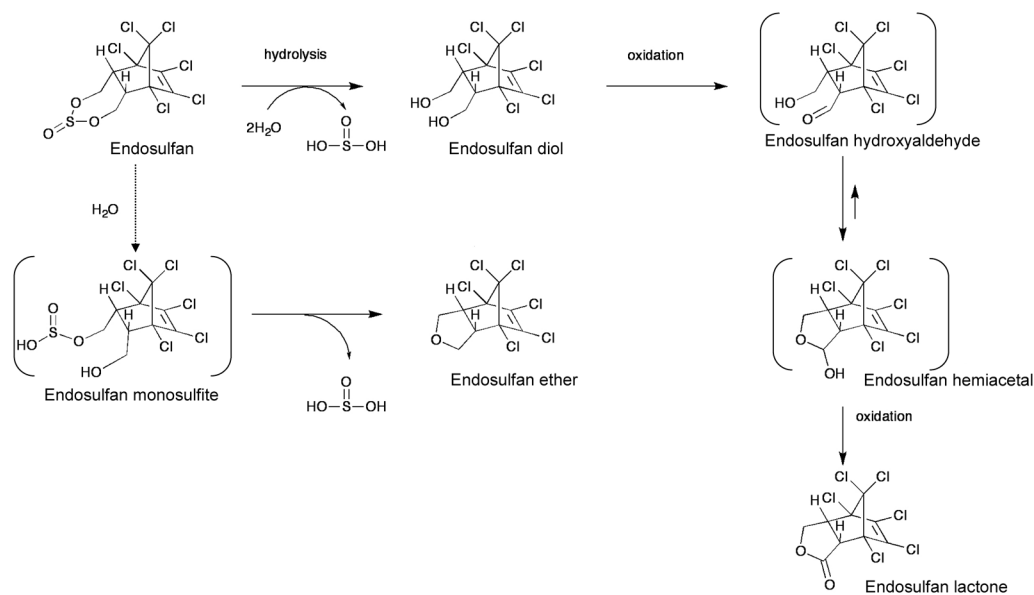


Fig. 6. Putative pathway of endosulfan degradation by strains W8 and Cm1-45. Compounds shown in parentheses are presumptive metabolites.

gregates that were obtained on a fine sieve following the washing procedure were aseptically transferred onto a plate of Martin agar medium and then incubated at 25°C for 5 days. The fungal colonies obtained were classified into several groups according to their morphology, growth rate and color, and representatives of each group were picked up with a sterile needle and sub-cultured on PDA.

The degradation experiment was performed using 35 strains of *Trichoderma* spp. and 133 strains (Kagoshima: 48, Ehime: 38, and Niigata: 47) of soil isolates. There was no endosulfan-degrading *Trichoderma* used in this study. However, two fungi (strains W8 and Cm1-45) isolated from soil D, which was contaminated with organochlorine pesticides, degraded α and β -endosulfan. Strain Cm1-45 degraded 47.2% of α -endosulfan and 25.1% of β -endosulfan during 14 days of incubation, while strain W8 degraded 53.3% of α -endosulfan and 11.1% of β -endosulfan. A time course degradation using strains W8 and Cm1-45 was then performed, during which both strains degraded more than 70 and 50% of α and β -endosulfan, respectively, over 28 days at 25°C (Fig. 5). However, only 3.2 and $1.2\ \mu\text{M}$ endosulfan sulfate was detected in the cultures of strains W8 and Cm1-45, respectively. Endosulfan sulfate as an initial substrate was not degraded by strains W8 and Cm1-45, and the recovery of this compound was found to be approximately 100%.

2.2. Identification of endosulfan-degrading fungi

The ITS rRNA gene sequences of strain W8 (501 nucleotides, GenBank accession no. AB542930) and Cm1-45 (501 nucleotides, GenBank accession no. AB542931) were compared with those of the fungal sequences in GenBank. Strains W8 and Cm1-45 exhibited high sequence similarities to those of *Mortierella* fungi. The highest sequence similarity (100%) was found with *Mortierella verticillata* (GenBank accession no. AY997063).

Strains W8 and Cm1-45 were closely related to *Mortierella verticillata*.

2.3. Detection of metabolites in endosulfan degradation

In the ethyl acetate extracts of cultures from strains W8 and Cm1-45, four different peaks with retention times of 6.57, 5.69, 7.93 and 9.86 min were observed upon GC/ECD analysis of α and β -endosulfan. These peaks were assigned as endosulfan diol, endosulfan ether, endosulfan lactone and endosulfan sulfate based on their having the same retention time as authentic standards. The time courses of endosulfan degradation and endosulfan diol production by strains W8 and Cm1-45 are shown in Fig. 5. When endosulfan was added to the MCD cultures, it decreased with accompanying growth of strains W8 and Cm1-45 and endosulfan diol was produced. In contrast, no production of endosulfan diol, endosulfan ether or endosulfan lactone was detected in the autoclaved control cultures. Endosulfan diol, endosulfan ether, endosulfan lactone and endosulfan sulfate generated molecular ion peaks at m/z 360, 342, 358 and 422, respectively, and the relative intensity of the chlorination pattern matched the isotope ratio of Cl_6 in all detected metabolites. The metabolites were detected at retention times of 15.1, 11.4, 13.3 and 15.8 min in endosulfan diol, endosulfan ether, endosulfan lactone and endosulfan sulfate, respectively, and these retention times and mass spectral patterns matched those of the individual standards. The proposed pathway of degradation by strain W8 or Cm1-45 is shown in Fig. 6.

Concluding remarks

Dieldrin- and endosulfan-degrading fungi were screened from soil contaminated with endosulfan and their degradation abilities were demonstrated. *Mucor racemosus* strain DDE decreased dieldrin by converting it to aldrin *trans*-diol phosphate *via* al-

drin *trans*-diol. *Mortierella* sp. strains W8 and Cm1-45 generate endosulfan diol as a first step in the degradation of endosulfan, then undergo further conversion to endosulfan lactone.

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